Ahmad Sebaq

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Summary

Senior Computer Vision & AI engineer (7 years) specializing in generative models, vision-language models, and agentic LLM pipelines. Technical lead who has shipped research to production and led teams of up to 8.

Education

Nile University — Giza, Egypt
M.S. in Communication and Information Technology, GPA: 4.0/4.0 — 2021–2026

Minia University — Minya, Egypt
B.S. in Biomedical Engineering, Grade: Very Good with Honors (84.6%) — 2014–2019

Experience

CyShield — New Cairo, Egypt
Senior Computer Vision Engineer *(Full-time)* — January 2025 – Current

- **Biometrics & Video Surveillance:** Built face and fingerprint 1:1 verification and 1:N watchlist identification for border control under strict latency and false-match constraints, and multi-camera stadium surveillance — detection, tracking, cross-view re-identification and action recognition for incident alerts on live streams.
- **Generative Models & VLMs:** Trained diffusion- and GAN-based inpainting models end to end (dataset curation, mask generation, LPIPS/FID/SSIM/PSNR evaluation) and fine-tuned vision-language models for grounding, captioning and VQA, with vector-search retrieval for RAG.
- **Technical Leadership:** Led a squad of 3 AI engineers — scoping, code review, mentoring, and aligning research direction with product deadlines.

CNTXT — Remote — UAE / Egypt
AI/Computer Vision Consultant *(Part-time)* — January 2026 – Current

- **Robotics Data Platform:** Built an egocentric data-collection and QA platform turning 100+ hours of 3-camera teleoperation video per week into LeRobot datasets (3D hand pose, automated video QA, VLM action annotation); auto-QA agreed with human reviewers on 90.8% of submissions.
- **VLM Fine-tuning (SFT + RL):** Fine-tuned Qwen3-VL-8B with QLoRA and a GRPO stage to segment and caption teleoperation video: segment-IoU F1 0.56 → 0.71, caption token-F1 0.22 → 0.57 on 497 held-out windows; adapted to a new client from 15 episodes.
- **Agentic RAG (Arabic):** Co-developed an Arabic citation-grounded RAG service on LangGraph, deployed on AWS EKS. Owned retrieval quality: Qwen3 embedding and reranking lifted Hit@1 0.84 → 0.95 and nDCG@10 0.85 → 0.93; human-calibrated LLM judges and a grader redesign lifted correct refusal 55% → 94%; latency cut from up to 22.6 s to ∼1.3 s.
- **GPU Serving, Speech & CV:** GPU inference services on AWS EKS/EC2 L4 (admission control, caching, SQS workers, S3 LoRA adapters); Arabic ASR benchmarked across 4 dialects; Mask2Former segmentation (mask mAP 0.59 → 0.68) and CLIP-based RGB/IR drone-frame matching.

Lumin Soft — Giza, Egypt
AI Team Lead *(Full-time)* — November 2023 – January 2025

- **Team Leadership:** Led a team of 8 AI developers — 1:1s, sprint planning, and technical mentorship.
- **Research-to-Production:** Delivered 3 production projects spanning segmentation, detection, tracking, re-identification and OCR, turning research prototypes into deadline-bound pipelines.
- **Sports Analytics (flagship):** Led a multi-model sports video pipeline — player/ball detection and tracking, team clustering, ReID, action recognition, possession tracking — with annotated video and metrics reports.

Applied Innovation Center Giza, Egypt
R&D Engineer *(Full-time)* October 2022 – November 2023
- Segmented land cover into 7 classes from Sentinel-2 and Planet imagery with U-Net and Transformer networks (98% overall accuracy across Egypt's main governorates), building end-to-end geospatial pipelines with GIS analysts.

Center for Informatics Science, Nile University Giza, Egypt
Research Assistant *(Full-time)* April 2021 – April 2023
- Classified rice crop diseases across 5 classes with ResNet and Transformer models, achieving 96% F1 on a 250K-image dataset.

Earlier Roles Egypt / USA
Computer Vision Engineer, part-time (TechnoIsland, VLens, Dawarha) 2019 – 2023
- Shipped national-ID OCR on AWS EC2, real-time recyclable detection on a Raspberry Pi (90 mAP, 11 FPS) and a microscopy-video detection/tracking pipeline (YOLOv3, Yolact, DeepSort); mentored student CV teams for Affectiva (2020).

Projects

BlacksmithAI *(Freelance, 2026)* blacksmithai.org
Natural-language to CAD agentic pipeline
- **Agentic pipeline:** Designed a multi-stage state machine (decompose → plan → assemble → execute → validate → repair) over a deterministic JSON IR with content-addressed caching, multi-modal image grounding, and human-in-the-loop review.
- **LLM tool-use & verification:** Built a Pydantic-validated tool catalog and pure verifier tools the LLM uses to self-check IR before execution; routed Claude Haiku and Gemini Flash via OpenRouter. *Stack:* FastAPI, Qdrant + BGE RAG, CadQuery, React + react-three-fiber, Docker.

TV Ad-Recognition Pipeline
Real-time broadcast ad detection via multi-modal fingerprinting
- Built a hybrid recognition system matching live TV broadcasts against an ad catalog using audio fingerprinting (Chromaprint), speaker embeddings (WavLM x-vectors), and scene-change keyframes (PySceneDetect + pHash + CLIP ViT-B/32 embeddings), fused over Elasticsearch kNN + BM25 search with NMS-suppressed airing events. Added a persistent-socket WebSocket endpoint for live per-channel recognition. *Stack:* two FastAPI services, Elasticsearch, PyTorch, FFmpeg; deployed as auto-start Windows services.

Sperm Analysis (CASA)
Computer-aided semen analysis desktop product
- **Analysis engine:** Built a CASA pipeline measuring sperm count, motility (per-cell kinematics via multi-object tracking), and morphology from microscopy and industrial-camera video — YOLO detection served through ONNX Runtime.
- **Product & deployment:** Shipped as an offline Windows desktop app — FastAPI backend (PyInstaller) wrapped in an Electron + React UI, industrial-camera capture (GenTL / Harvesters), encrypted model weights, HTTPS licensing service, and PDF clinical reports.

Research

Fatima Fellowship Stanford, USA
Research Intern *(Part-time)* September 2023 – August 2024
- Developed generative models to enhance low-resolution MRI images, focusing on model optimization and validation for clinical applications.

Linköping University Linköping, Sweden
Research Intern *(Part-time)* October 2022 – January 2023
- Developed GANs to manipulate Maxar aerial images by adding realistic seasonal effects, collaborating with geospatial analysts to ensure accuracy.

Publications

[1] T. A. Abdel Salam, A. M. Abdelsalam, N. R. Sholkamy, and **Sebaq, Ahmad**, "Assessment of artificial intelligence in detecting errors on panoramic radiographs", *Imaging Science in Dentistry*, vol. 56, e12, 2026. DOI: 10.5624/isd.20250242

[2] **Sebaq, Ahmad** and M. ElHelw, "Rsdiff: Remote sensing image generation from text using diffusion model", *Neural Computing and Applications*, vol. 36, no. 36, pp. 23 103–23 111, 2024. DOI: 10.1007/s00521-024-10363-3

[3] Y. A. Alnaggar, **Sebaq, Ahmad**, K. Amer, E. Naeem, and M. Elhelw, "Rice plant disease detection and diagnosis using deep convolutional neural networks and multispectral imaging", in *International Conference on Model and Data Engineering*, Springer, 2023, pp. 16–25. DOI: 10.1007/978-3-031-21595-7_2

[4] M. S. M. Bahgat and **Sebaq, A. M.**, "An Analytical Computational Algorithm for Solving a System of Multipantograph DDEs Using Laplace Variational Iteration Algorithm", in *Advances in Astronomy*, Jun. 2021, p. 16. DOI: 10.1155/2021/7741166

Awards

- Full Master Scholarship from Nile University 2021–2023
- 3rd Place, Global Monitoring for Environment and Security (GMES) & Africa Incubation Challenge 2021
- 3rd Place in Affectiva Makeathon with Inside Out project 2020
- Nokia Africathon Finalist with ALS project 2019

Skills

- **Programming:** Python, C++, JavaScript / TypeScript, React
- **Generative AI & LLMs:** LLM orchestration (LangGraph, LangChain), multi-agent pipelines, tool-use / function-calling, agentic RAG, multi-modal VLMs, local inference (llama-cpp-python), vector & hybrid search (Qdrant, Elasticsearch; dense + BM25, RRF fusion, cross-encoder reranking, ColBERT late interaction), LLM-as-judge evaluation, prompt engineering
- **Computer Vision:** OpenCV; segmentation (U-Net, Mask R-CNN, Swin), detection & tracking (YOLO, Yolact, DeepSort), OCR, re-identification, 3D hand-pose / SLAM, CLIP cross-modal matching
- **ML / Deep Learning:** PyTorch, ONNX Runtime, HuggingFace, scikit-learn; CNNs, Transformers, diffusion models, GANs; ASR (Whisper), speaker diarization (Pyannote), denoising (DeepFilterNet3), audio fingerprinting
- **MLOps & Deployment:** FastAPI, Docker, Jenkins CI, AWS (EC2), Git; Postgres + SQLAlchemy/Alembic, JWT auth, structured logging & Sentry; desktop packaging (Electron, PyInstaller); edge deployment (Jetson Nano & Orin, Raspberry Pi)
- **Robotics & Geospatial:** LeRobot imitation-learning data pipelines, industrial-camera capture; QGIS, Sentinel-2 / Planet / Maxar satellite imagery